

In his book [Colbrook 26] Matt introduces a very interesting algorithm to compute the SPECTRA and more interestingly the PSEUDOSPECTRA of differential operator. Yet the functional analytic setting remains mysterious to me, and in this note I'll try to explain why.

Matt's idea is to resort to operator folding so we consider a closed operator A densely defined on our Hilbert space H and since we want to compute its eigenvalues this operator should be an ENDOMORPHISM so let $A: H \rightarrow H$.

EXAMPLE The classical operator that satisfies

$$\text{Matt's hypothesis is } \Delta: H_0^2 \longrightarrow L^2.$$

$$\quad \quad \quad \text{ii}$$

$$\quad \quad \quad H_0^1 \cap H^2$$

Then Matt says:

"Under suitable conditions on $D(A)$ and $D(A^*)$ " we consider

$$\Phi(z, A) := \max \{ \sigma_{\text{inf}}(A - zI), \sigma_{\text{inf}}(A - zI)^* \}$$

①

where $\sigma_{\inf}$ are given by the quadratic forms associated with the resolvent and its adjoint, i.e.

$$\sigma_{\inf}(A - zI) = \sqrt{\inf_{f \in D(A)} R^{(1)}(f)}$$

$$\sigma_{\inf}(A - zI) = \sqrt{\inf_{f \in D(A^*)} R^{(2)}(f)}$$

with

$$R^{(1)}(f) := \frac{\|(A - zI)f\|_H^2}{\|f\|_H^2}$$

$$R^{(2)}(f) := \frac{\|(A - zI)^*f\|_H^2}{\|f\|_H^2}.$$

So far I'm happy and everything appears to be formal my instead arise when we comment below equation 3.26, i.e. that this reduces computing the PSEUDOSPECTRA to computing the eigenvalue of the SELF-ADJOINT operator,
 $(A - zI)^* (A - zI).$

So my understanding is that this is the normal operator associated with $(A - zI)$. Now following [KATO, PERTURBATION THEORY FOR LINEAR OPERATORS 9.275] this operator is not well posed as a densely-defined closed operator if we take as domain the domain $D(T)$ but we need to define a special domain. Not only that but if I'm interested in applying Bobusko - Osborn approximation theory for FEM to $(A - zI)^*(A - zI)$ I can't because this is not a bounded operator and it doesn't have a compact inverse. If instead of considering $(A - zI)^*(A - zI)$ > KATO's normal operator we assume it is equal to the product of two operators, i.e. $(A - zI)$ and its adjoint $(A - zI)^*$, then I believe I (3)

have a primal dual error since,
the Banach adjoint is defined as

$$(A - zI)^*: H^* \longrightarrow H^* \text{ and}$$

then we would have:

$$H \xrightarrow{(A - zI)} H \quad \underline{\underline{\times}} \quad H^* \xrightarrow{(A - zI)^*} H^*$$

A standard way to resolve this issue,
in particular if H is L^2 is to
silently identify H with its dual,
which I'm not a big fan of but
then we have:

$$H \xrightarrow{(A - zI)} H \cong H^* \xrightarrow{(A - zI)^*} H^* \cong H$$

Notice that this coarsens the composition
to remain an endomorphism and
prevent the primal dual-error.
But due to the silent $L^2 \cong (L^2)^*$
identification the identity considers (h)

with the identity and the correct form of the operator would be

$$M_{L^2}^{-1} (A - zM_{(L^2)})^* \quad M_{L^2} \quad (A - zM) \quad (*)$$

$$H \xrightarrow{(A - zM_{L^2})^*} H \xrightarrow{M} H^* \xrightarrow{(A - zM) M^{-1}} H^* \rightarrow H$$

The issue is that this operator doesn't land well in finite element space, the first issue being that FEM operators are weakly defined since they are variational formulations, so usually they go from $A \rightarrow H^*$, and also I don't see how to read a variational formulation from $(*)$.

A PATH THAT MIGHT WORK

A path that functional analytically make sense for me is the following (which I show case with a concrete example). Let $V \subset H$ in particular $H = L^2$ and $V = H \cap \text{div}(0)$ then I have a differential operator of the first order:

$$A: V \times V \rightarrow V^* \times V^*$$

$$\langle A \begin{pmatrix} E \\ H \end{pmatrix}, \begin{pmatrix} G \\ F \end{pmatrix} \rangle_{V^* \times V^*} = \langle -i \bar{\nabla} \times E, F \rangle_{L^2} + \langle i \hat{\nabla} \times H, G \rangle_{L^2}$$

effectively since this is a first order operator it goes not into the dual of V but in the one of H which make sense since

$H^* \subset V^*$. Then we can consider the Riesz map $M: H^* \rightarrow H$ and write the folded operator (6)

$$(A - zM)^* M^{-1} (A - zM) \quad (**)$$

$$V \times V \xrightarrow{A - zM} H^* \times H^* \subseteq V^* \times V^*$$

$$\downarrow M^{-1}$$

$$\left. \begin{array}{l} \text{Canonical} \\ \text{double} \\ \text{dual} \\ \text{identification} \\ \text{numerically} \end{array} \right\} \longrightarrow \begin{array}{c} H \times H \\ \text{SI} \\ (H^* \times H^*)^* \end{array} \xrightarrow{(A - zM)^*} V^* \times V^*$$

this is an identity

This now has a variational form associated to it, i.e.

$$\begin{aligned} & [(\bar{U} \times E, \bar{V} \times F) + (\hat{U} \times H, \hat{V} \times G) - (\bar{z} + z) : (\hat{V} \times H, F) \\ & + (\bar{z} + z) : (\bar{V} \times E, G) + \bar{z} z (E, F) \\ & + \bar{z} z (H, G)] \\ & \hookrightarrow a(E, F, H, G) \end{aligned}$$

Furthermore due to elliptic regularity we know that the solution of the variational problem associated ⑦

with this operator live in a subspace $W \subset V$ and therefore instead of studying the eigenvalue of (xx) we can study BoFFI style the eigenvalue of the solution operator $T: V \times V \rightarrow W \times W$

$$a(T(f,g)_1, F; T(f,g)_2, G) = (f, F), (g, G)$$

and extend T to $T: V \times V \rightarrow V \times V \quad \forall F, G \in V \times V$
 which is a compact endomorphism by $W \times W \subset V \times V$.